RAMCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Academic Year: 2019 - 2020 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: CS8691 Artificial Intelligence

Type of Activity: Collaborative Learning

Topic: Problems on Unification, Resolution

Date: 18-02-20

Objectives:

O1: To enable the students to learn and practice the problems on unification and resolution.

Justification for chosen the topic:

Problems on the topic unification and resolution were important and should be practiced. For better understanding of the following topics the collaborative tutorial was conducted for solving different problems.

Activity Description:

Collaborative learning is an educational approach to teaching and learning that involves groups of students working together to solve a problem. The students were divided into group based on their own interest. The topic for the tutorial was announced to the students through a canvas and classroom. The team members work together to solve a problem. The tutorial given to the student was as attached in annexure-I.

<u>**Reflection critique**</u>

Pre-implementation:

Identified Challenges:

- Chances of few students to deviate from the plan.
- Instructed to bring tutorial note and relevant material.

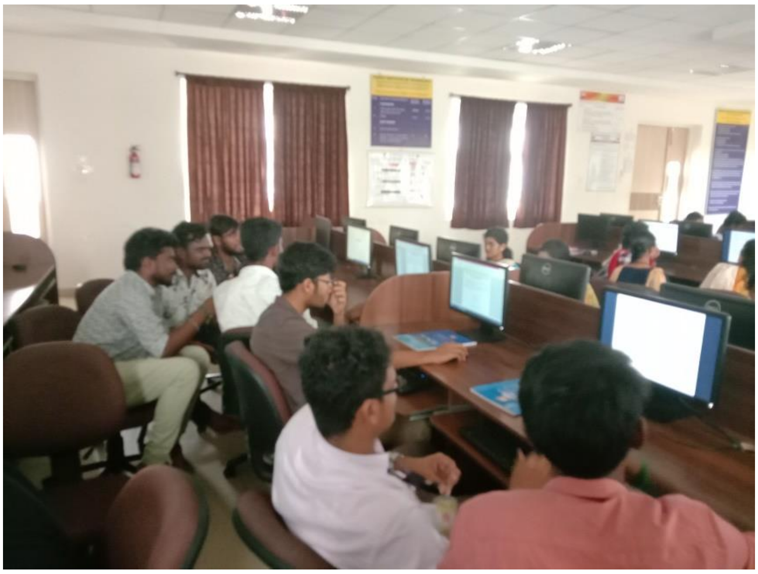
Post Implementation:

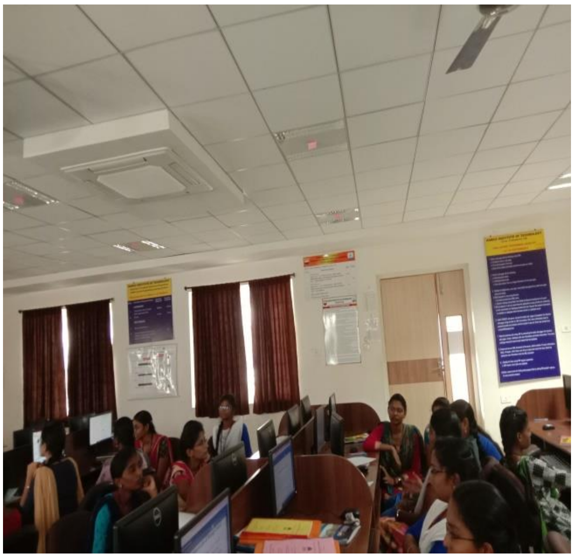
- Leadership Quality of the student improved.
- Students actively participate and this activity encouraged the students to share their knowledge with others.
- With the actions taken initially, the activity is carried out smoothly with maximum Commitment from students.

Addressing Challenges:

- Addressed the importance of steps to convert the predicate logic.
- Some of the students made mistake on the step reduce negation that was addressed with related problems.

Glimpses





Relevance of Program Outcomes

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	3	2	1		1			2		2		3

References:

https://www.gdrc.org/kmgmt/c-learn/
https://www.edutopia.org/topic/collaborative-learning
https://www.ritrjpm.ac.in/images/computer-science/28.CS8501_ColloborativeLearning.pdf

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Annexure-1

RAMCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Academic Year: 2019-2020 (Even Semester)

TUTORIAL SHEET

Degree, Semester& Branch: VI Semester B.E. CSE
Course Code & Title: CS8691 Artificial Intelligence
Name of the Faculty member: Mrs.S.Manjula
Tutorial No.: 2 **Date: 18-02-2020**

Part A

1. Convert the following sentences to predicate logic, clause form and thus prove by resolution that- "Marcus was a Roman". [CO3, L3]
    - Marcus was a Pompeian.
    - All Pompeian were Romans.
2. Convert the following sentences to clausal form. [CO3, L3]
    - $p \wedge q \Rightarrow r \vee s$
    - $p \vee q \Rightarrow r \vee s$
    - $\neg (p \vee q \vee r)$
    - $p \wedge q \Leftrightarrow r$
3. Represent the following in semantic net, "All students like the subject AI". [CO3, L3]
4. Discover the operation of the unification algorithm on each of the following pairs of literals: A) f(Marcus)and f(Caesar) B) f(x) and f(g(y)) C) f(Marcus, g(x, y))and f(x, g(Caesar, Marcus)) [CO3, L3]
5. What is refutation principle? [CO3, L1]

Part B

1. Turn the following sentences into formulae in first order predicate logic. Convert FOL into clause form [CO3, L3]
    - John likes all kinds of food.
    - Apples are food.
    - Chicken is food.
    - Anything anyone eats and isn't killed by is food.
    - Bill eats peanuts and is still alive.
    - Sue eats everything Bill eats.
2. From "Horses are animals," it follows that "The head of a horse is the head of an animal. "Demonstrate that this inference is valid by carrying out the following steps: [CO3, L3]
    a. Translate the premise and the conclusion into the language of first-order logic. Use three predicates: HeadOf (h,x), Horse(x), and Animal(x).
    b. Negate the conclusion, and convert the premise and the negated conclusion into conjunctive normal form.
    c. Use resolution to show that the conclusion follows from the premise.

CO3: The students will be able to represent a problem using first order and predicate logic.

L1: Remember; L2: Understand; L3: Apply